

CE307 Advanced Java Programming

Date: 04.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q – I (a) 1. Which type of Statement can execute parameterized queries? [07]
- A. PreparedStatement
 - B. ParameterizedStatement
 - C. ParameterizedStatement and CallableStatement
 - D. All kinds of Statements (i.e. which implement a sub interface of Statement)
2. What is correct about DDL statements (create, grant...)?
- A. DDL statements are treated as normal SQL statements, and are executed by calling the execute() method on a Statement (or a sub interface thereof) object
 - B. To execute DDL statements, you have to install additional support files
 - C. DDL statements cannot be executed by making use of JDBC, you should use the native database tools for this.
 - D. Support for DDL statements will be a feature of a future release of JDBC.
3. If a DatagramSocket constructor fails to set up a DatagramSocket properly, an exception of type _____ occurs.
- A. SocketException
 - B. CloseException
 - C. BindException
 - D. IOException
4. Method getLocalHost returns a(n) _____ object containing the local IP address of the computer on which the program is executing.
- A. InetAddress
 - B. Socket
 - C. ServerSocket
 - D. DatagramSocket
5. Which of the following is a marker interface that does not have any methods of its own?
- A. DatabaseMetaData
 - B. ResultSetMetaData
 - C. Remote
 - D. SocketOptions
6. What is the default port of RMI registry?
- A. 80
 - B. 99
 - C. 1099
 - D. 8080
7. Which of following exception is not related to Socket class?
- A. BindException
 - B. ConnectException
 - C. ProtocolException
 - D. None of the above

- Q - 2 (a) How does RMI provide abstraction over socket? How do RMI clients contact remote contact remote RMI server? List out all the steps. [04]
- (b) What is InetAddress? List and explain the factory methods available in InetAddress? [04]
- (c) What is MulticastSocket? Explain how user can join the multicast group. [03]

OR

- Q - 2 (a) Write a RMI client and server program to find the binary equivalent of the integer number passed by the client program. Use an exception handling mechanism to handle float errors. [07]
- (b) What is the difference between Server and ServerSocket? Explain the accept() and bind() methods of ServerSocket. [04]
- (c) Why there are two separate classes for UDP in Java (e.g. DatagramPacket and DatagramSocket)? List the methods to set data and length of data for the packet. [03]
- Q - 3 (a) Assume an employee table with fields like empno, name and phone no fields. Write a database application which allows inserting, updating and deleting records from table using JDBC. Print information of all employees in alphabetical order. [06]
- (b) What is JDBC? Compare the type 1 and type 4 JDBC driver. [04]
- (c) Write the difference between PreparedStatement and CallableStatement. How SQL can be executed in batch? Explain with suitable example. [04]

OR

- Q - 3 (a) Consider a Bank table with attribute Account No, Customer Name, Balance, Phone and Address. Write a database application which allows insertion, updation and deletion of record in the Bank table. Print values of all customers whose balance is greater than 2,000. [06]
- (b) What are ResultSetMetaData and DatabaseMetaData? Explain the purpose of DriverManager in jdbc. [04]
- (c) Explain the constants related to ResultSet which are used to initialize connection with the database. [04]

SECTION – II

- Q - 4 1. What is legal about JSP scriptlets? [07]
- loop can begin in one Scriptlet and end in another
 - Statements in Scriptlets should follow Java Syntax
 - Semicolon is needed at the end of each statement in a Scriptlet
 - All the above
2. In JSP, how can you know what HTTP method (GET or POST) is used by client request?
- by using request.getMethod()
 - by using response.getMethod()
 - by using config.getMethod()
 - impossible to know
3. Which method is called first time when a Servlet is invoked?
- start()
 - run()
 - service()
 - init()
4. Servlet may not have a main method. State true or false. Justify your answer.
5. Which of the following is proper sequence of objects created by servlet container during life cycle of application?
- ServletContext, ServletConfig, HttpServletRequest, HttpServletResponse
 - ServletConfig, HttpServletResponse, HttpServletRequest, ServletContext
 - ServletContext, HttpServletRequest, HttpServletResponse, ServletConfig
 - ServletConfig, HttpServletRequest, HttpServletResponse, ServletContext
6. Context attributes need to be synchronized in servlet. State true or false. Justify your answer.
7. Once session is destroyed, calling any method on HttpSession generates _____ exception.

- 5 (a) Write a program using Servlet, JSP, JDBC, and XML for developing an online application for the shopping of computer science books. (Hint: use concept of session tracking) You have to create a database for book title, author(s) of book, publisher, year of publication, price. Make necessary assumptions for book shopping. [07]
- (b) What is deployment descriptor? Write a tag in deployment descriptor that is used to handle the error generated by web container. [04]
- (c) Write the differences between servlet context and servlet config. Write methods name to get reference of each object in servlet. [03]

OR

- Q - 5 (a) Using Servlet, JSP, JDBC, and XML, create a Web application for a recruitment agency to providing assistance in searching the candidates from its databases as per requirements of its various clients. Make necessary assumptions while developing this application. [07]
- (b) What is response send redirect and request dispatcher? Which and why exception may be thrown by them? [04]
- (c) Write differences between GET and POST methods of HTTP. [03]

Q - 6

- (a) What is session? What are the possible ways to manage session in servlet? Explain the listeners related to HttpSession. [06]
- (b) What are the three different types of elements in JSP? Which are advantages of JSP over Servlet? List the implicit objects of JSP. [06]

OR

- (b) Explain the architecture of J2EE. Show how the components of J2EE fit in Model View Controller (MVC) architecture. [06]
- (c) What is tag library? List the steps to create a tag library in JSP. [02]
